

Juan David Munoz Bolanos

Optical Systems Engineer

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PROFESSIONAL SUMMARY

Optical Systems Engineer specializing in high precision metrology, optical modeling, and opto-mechatronics. Delivering successful peer-reviewed projects, including integration of adaptive optics for diffraction-limited multi-photon microscopy and full-cycle development of 1064 nm fiber probe Raman spectrometers. ASML Summer School participant driven to contribute to state-of-the-art optical systems and yield alignment technologies.

TECHNICAL COMPETENCIES

High Precision Metrology & Simulation

Deep knowledge of imaging physics, custom Python imaging simulations (Zernike), and optical modeling. Familiar with Zemax OpticStudio for tolerancing custom dispersive probes and resolving complex phase aberrations.

Opto-Mechatronics & Automation

Proven ability to automate optical benchtop experiments and integrate hardware with software, including synchronizing fluidic tuneable lenses via custom Python, C++, and LabVIEW interfaces.

Machine Vision & Image Processing

Industrial deployment of machine vision pipelines (Halcon, OpenCV) for high-speed automated inspection, yield assessment, and defect detection.

System Integration & Architecture

End-to-end R&D experience bringing complex optical setups from concept to fully functional, calibrated prototypes in both academic and industrial settings.

PROFESSIONAL EXPERIENCE

Medical University of Innsbruck

PhD Researcher - Optical Metrology & Precision Measurements

Innsbruck, Austria

Jan 2022 – Present

- **Adaptive Optics Integration (Biomedical Optics Express '23 / ACS Photonics '25):** Designed an automated wavefront sensing system using a custom C++/Python/LabVIEW control stack. Synchronized 63-actuator Deformable Phase Plates (DPP) and Texas Instruments Phase Light Modulators (TI-PLM) to correct aberrations, achieving diffraction-limited resolution in deep-tissue microscopy.
- **Interferometric Phase Retrieval & Modeling:** Engineered system-level wavefront sensing frameworks using F-SHARP interferometry and batched-DASH optimization. Performed optical simulations alongside Python Zernike decomposition, experimentally resolving spatial shifts via a novel "3N algorithm."
- **Project Management:** Managed the specification and procurement of high-precision opto-mechanical components, interferometers, and DAQ electronics, steering complex optofluidic setups to peer-reviewed completion.

Leibniz Institute of Photonic Technology (IPHT)

R&D Researcher (Photonics & Data)

Jena, Germany

Jun 2020 – Dec 2021

- **Custom Spectrometer Prototyping (Applied Sciences '24):** Led the optical design, tolerancing, and construction of a dispersive 1064 nm fiber probe Raman spectrometer for clinical applications, handling rigorous detector alignment and calibration.
- **Data Analysis & AI:** Applied supervised machine learning to 2D hyperspectral Raman images from the custom probe, achieving high-accuracy tissue classification and accelerating characterizations.

INTECOL S.A.S.

Medellín, Colombia

- **Industrial Vision Deployment:** Deployed Halcon-based image processing pipelines for high-speed manufacturing inspection, resolving production failures through custom illumination and optics.
- **System Architecture:** Selected cameras, sensors, and illumination schemes to fully implement and validate optical inspection setups on active production lines.

EDUCATION

Medical University of Innsbruck

Austria

Ph.D. in Optical Systems Engineering (Adaptive Optics & Microscopy)

Jan 2022 – Present

Friedrich Schiller University Jena

Germany

M.Sc. in Photonics

2019 – 2021

University of Cauca

Colombia

B.Sc. in Physics Engineering

2012 – 2018

PUBLICATIONS & AWARDS

- **Awards and Selections:** Selected for ASML and ZEISS Summer Schools; awarded COLFUTURO Grant 2026 for best Colombian professionals.
- 'Juan David Munoz Bolanos et al. Confocal Raman Microscopy with Adaptive Optics. ACS Photonics 12(1), 176–184 (2025).
textitWork Done': 'Integrated adaptive optics to achieve diffraction-limited resolution in deep-tissue imaging, resolving phase aberrations experimentally.'
- 'Juan David Munoz Bolanos et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer. Applied Sciences, 14(11), 4726 (2024).
textitWork Done': 'Concepted and developed a custom dispersive spectrometer probe, covering full optical design, tolerancing, and detector calibration.'
- 'Sohmen M., Juan David Munoz Bolanos et al. Optofluidic adaptive optics in multi-photon microscopy. Biomedical Optics Express 14(4), 1562–1578 (2023).
textitWork Done': 'Integrated tuneable lenses into a scanning microscope and built the software interface for fast, aberration-free focusing.'

LANGUAGES

English (Fluent / Professional),
German (B2 - Working Proficiency),
Spanish (Native)

REFERENCES

Prof. Monika Ritsch-Marte

Medical University of Innsbruck

Head of Institute for Biomedical Physics

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Dr. Christoph Krafft

Leibniz Institute of Photonic Technology

Group leader for Spectroscopy Imaging

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